ECE 330 · LECTURE 2

Single-Phase AC Circuits, Power, and Phasors

From a physical load model to active power, reactive power, power factor, and a compact frequency-domain solution

THE TWO QUESTIONS For a load connected to an AC source: (1) What current flows? (2) How much average power is converted into useful output? These questions are related, but they are not identical.

Learning objectives

- Model a linear single-phase AC load using R, L, and C elements.
- Distinguish transient response from sinusoidal steady-state response.
- Derive instantaneous, active, reactive, and apparent power.
- Interpret power factor as utilization of voltage and current capacity.
- Use phasors to replace a differential equation with complex algebra.

Suggested reading: Sections 2.1–2.3 (Pai).

1. Element models: two equivalent views

Linear circuit elements can be described in the time domain or, after restricting attention to sinusoidal steady state at angular frequency ω, by algebraic relations between voltage and current phasors.

Element	Time-domain relation	Sinusoidal steady-state relation
Resistance, R	$v(t) = R\, i(t)$	$V = R\, I$
Inductance, L	$v(t) = L\, di(t)/dt$	$V = j\omega L\, I$
Capacitance, C	$i(t) = C\, dv(t)/dt$	$V = [1/(j\omega C)]\, I$

KEY IDEA In sinusoidal steady state, differentiation becomes multiplication by jω. This is the mathematical reason phasors simplify AC circuit analysis.

2. A motivating model: a slightly inductive heating element

Consider a heating appliance connected to an AC wall outlet. The resistance R represents conversion of electrical energy into heat. The inductance L represents the effect of the coiled conductor and associated magnetic field.

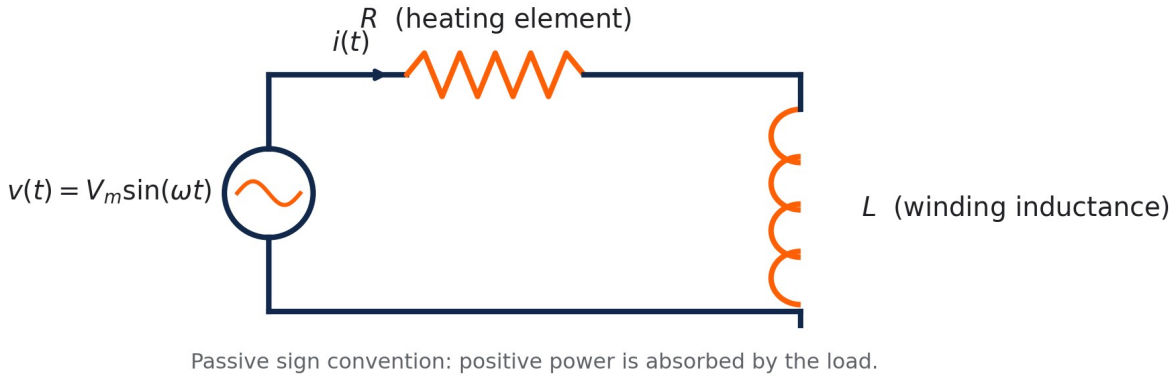


Figure 1. Series RL model of a heating load connected to a sinusoidal source.

MODELING NOTE An actual household toaster is nearly resistive and normally operates close to unity power factor. The inductance is retained here—and conceptually exaggerated—to expose behavior that becomes important in motors, transformers, and other AC loads.

2.1 Time-domain equation

Applying Kirchhoff's voltage law with the passive sign convention gives

$$Ri(t) + L\frac{di(t)}{dt} = v(t) = V_m \sin(\omega t)$$

The complete solution contains a natural response that decays with time constant $\tau = L/R$ and a forced sinusoidal response. After the natural response decays, the circuit is in sinusoidal steady state. In that regime, both voltage and current repeat every period $T = 2\pi/\omega$.

For the North American grid, $f = 60$ Hz and $\omega = 2\pi f$. For many other power systems, $f = 50$ Hz.

2.2 Steady-state current

The forced steady-state current has the same frequency as the source, but its magnitude is reduced and its phase is shifted:

$$i(t) = I_m \sin(\omega t - \phi)$$

$$I_m = \frac{V_m}{\sqrt{R^2 + (\omega L)^2}}, \qquad \phi = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

Because L > 0 gives φ > 0, the current reaches corresponding points in its cycle φ/ω seconds after the voltage. We therefore say that current lags voltage by the angle φ.

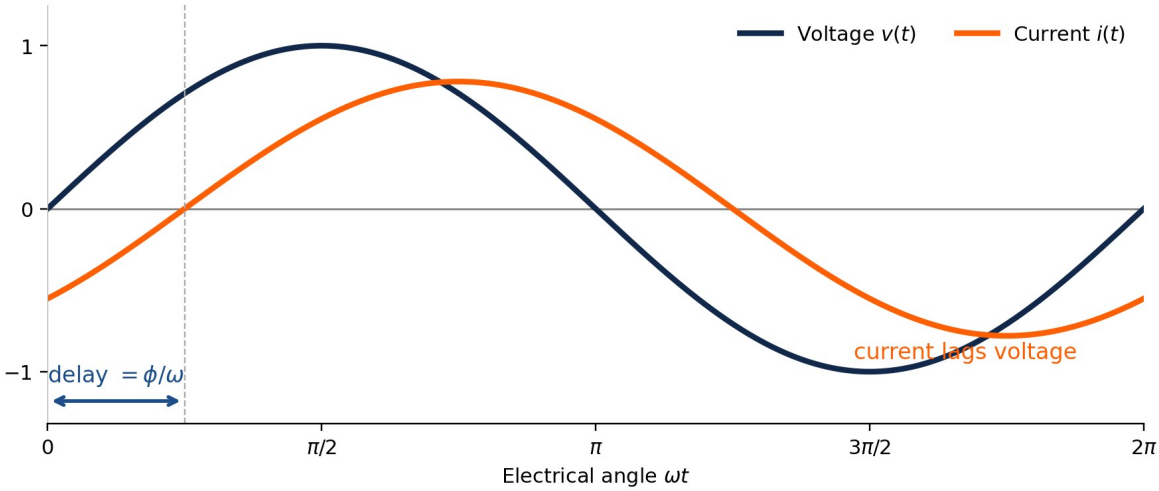


Figure 2. Voltage and current in a series RL load. Current lags voltage by φ radians, corresponding to a time delay φ/ω.

3. Instantaneous power

With the indicated voltage polarity and current direction, the instantaneous power absorbed by the load is

$$p(t) = v(t)i(t)$$

Substituting the voltage and current waveforms and applying a product-to-sum identity yields

$$p(t) = \frac{V_m I_m}{2}[\cos\phi - \cos(2\omega t - \phi)]$$

Expanding the second cosine separates the result into a constant term and two terms that oscillate at twice the source frequency:

$$p(t) = \frac{V_m I_m}{2}\cos\phi\,[1 - \cos(2\omega t)] - \frac{V_m I_m}{2}\sin\phi\sin(2\omega t)$$

3.1 Active and reactive power

Define active power P and reactive power Q as

$$P = \frac{V_m I_m}{2}\cos\phi, \qquad Q = \frac{V_m I_m}{2}\sin\phi$$

Then the instantaneous power can be written compactly as

$$p(t) = P[1 - \cos(2\omega t)] - Q\sin(2\omega t)$$

Averaging over one fundamental period removes both oscillatory terms:

$$\langle p(t)\rangle = \frac{1}{T}\int_0^T p(t)\,dt = P$$

PHYSICAL MEANING P is the average rate of net energy conversion—heat, mechanical output, light, or another useful form. Q measures reversible energy exchange with electric or magnetic fields. Reactive power is not net energy consumption, but it still requires current-carrying capacity.

3.2 Two limiting cases

The limiting cases reveal the physics more clearly than the general expression alone.

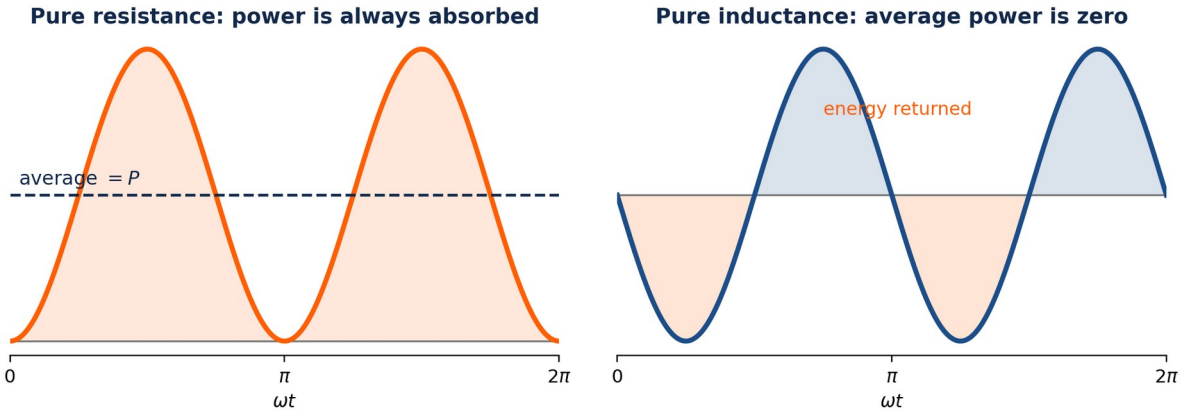


Figure 3. Instantaneous power for pure resistance and pure inductance. In both cases, p(t) varies at twice the source frequency.

Pure resistance (L = 0): φ = 0°, Q = 0, and p(t) = P[1 − cos(2ωt)]. Voltage and current are in phase; the instantaneous power is never negative.

Pure inductance (R = 0): φ = 90°, P = 0, and p(t) = −Q sin(2ωt). The inductor alternately stores energy in its magnetic field and returns it to the source, giving zero average power in the ideal case.

4. RMS values and the power triangle

Power-system voltage and current ratings are normally expressed using root-mean-square (RMS) values. For any periodic waveform x(t),

$$X_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T x^2(t)\,dt}$$

For a sinusoid, the RMS value is the peak value divided by √2:

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}}, \qquad I_{\text{rms}} = \frac{I_m}{\sqrt{2}}$$

Therefore the peak-value expressions above become

$$P = V_{\text{rms}} I_{\text{rms}} \cos\phi \ [\text{W}]$$

$$Q = V_{\text{rms}} I_{\text{rms}} \sin\phi \ [\text{var}]$$

$$S = V_{\text{rms}} I_{\text{rms}} \ [\text{VA}]$$

The apparent power S is the product of the available RMS voltage and current capacity. For the sinusoidal case,

$$P^2 + Q^2 = S^2$$

$$S^2 = P^2 + Q^2, \qquad \text{pf} = P/S = \cos\phi$$

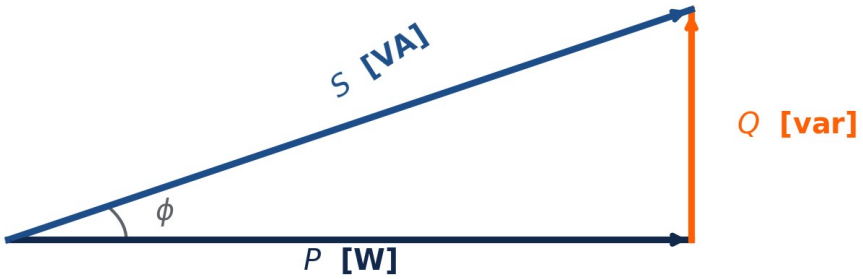


Figure 4. Power triangle for an inductive load. Positive Q is shown upward by convention.

5. Power factor and infrastructure utilization

For the sinusoidal linear load considered here, the power factor is

$$\text{pf} = \frac{P}{S} = \cos\phi$$

Power factor quantifies how effectively a given voltage and current capacity delivers average power. Wiring, transformers, switches, and protection devices must be sized for RMS voltage and RMS current—not only for active power.

NUMERICAL EXAMPLE Delivering 1.00 kW from a 120 V RMS source requires 8.33 A at unity power factor, but 16.67 A at pf = 0.50. For the same active power, the lower-power-factor load doubles the current and, for fixed wiring resistance, quadruples I^2R loss.

An inductive load has lagging power factor because its current lags its voltage. A capacitive load has leading power factor. The simple identity pf = cos φ applies here because both voltage and current are sinusoidal; nonlinear loads require a more general definition.

6. Phasors: replacing differential equations with algebra

A phasor encodes the RMS magnitude and phase of a single-frequency sinusoid. Using the sine reference adopted in these notes,

$$x(t) = \sqrt{2}\,X_{\text{rms}}\sin(\omega t + \alpha) \quad \Longleftrightarrow \quad \widetilde{X} = X_{\text{rms}}\angle\alpha$$

IMPORTANT DISTINCTION A phasor is not the full Fourier transform of a signal. It is a compact magnitude-and-phase representation of one sinusoidal steady-state component.

For the RL circuit, differentiation maps to multiplication by jω. The time-domain differential equation therefore becomes

$$(R + j\omega L)\tilde{I} = \widetilde{V}$$

Choosing the voltage as the phase reference gives

$$\widetilde{V} = V_{\text{rms}}\angle 0^\circ$$

$$\tilde{I} = \frac{\tilde{V}}{R + j\omega L} = \frac{V_{\text{rms}}}{\sqrt{R^2 + (\omega L)^2}} \angle \left[-\tan^{-1}\left(\frac{\omega L}{R}\right) \right]$$

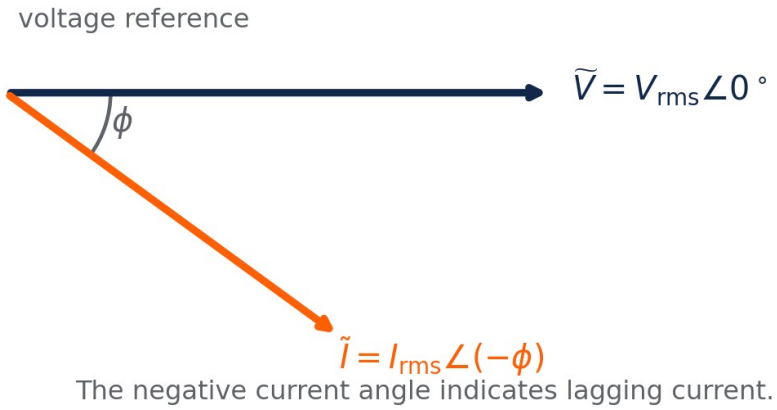


Figure 5. Voltage and current phasors for an inductive load. The current angle is -φ, so current lags voltage by φ.

THE PAYOFF The time-domain differential equation produced the physical picture. The phasor method produces the same steady-state magnitude and phase using complex algebra—without repeatedly solving differential equations.

7. Summary

Current phase. A series RL load draws a sinusoidal current that lags the source voltage.

Instantaneous power. It contains a constant term and oscillatory terms at 2ω.

P and Q. Active power is the average rate of net energy conversion; reactive power describes reversible field-energy exchange.

Apparent power. S represents RMS voltage-current capacity, with $S^2 = P^2 + Q^2$ for sinusoidal operation.

Power factor. It indicates how much of the available VA capacity becomes average power.

Phasors. They convert sinusoidal steady-state differential equations into algebraic equations involving complex impedances.

Essential relationships

Topic	Essential relationship
RL current	$I_m = V_m / \sqrt{[R^2 + (\omega L)^2]}, \quad \varphi = \tan^{-1}(\omega L/R)$
Instantaneous power	$p(t) = P[1 - \cos(2\omega t)] - Q \sin(2\omega t)$
RMS power	$P = VI \cos\varphi, \quad Q = VI \sin\varphi, \quad S = VI \quad (V \text{ and } I \text{ are RMS})$
Power triangle	$S^2 = P^2 + Q^2, \quad pf = P/S = \cos\varphi$
Phasor current	$\tilde{I} = \tilde{V}/(R + j\omega L) = I\angle(-\varphi)$

Check your understanding

Q1. Why does instantaneous power oscillate at twice the source frequency?

Q2. Can a load draw substantial RMS current while consuming zero average power? Explain using an ideal inductor.

Q3. A 120 V RMS load consumes 900 W at power factor 0.75 lagging. Determine the RMS current and apparent power.

Q4. What conceptual step allows the differential operator d/dt to become multiplication by jω?

ANSWER CHECK Q1: Multiplying two sinusoids produces a component at 2ω. Q2: Yes; an ideal inductor can carry current while its average power remains zero. Q3: I = 10 A RMS and S = 1.20 kVA. Q4: Restricting the analysis to a single-frequency sinusoidal steady state allows d/dt → jω.